Analyzing the Factors of Unaccounted-for Water in a Costal Municipal Potable Water System: A Case Study of Cape Coral, Florida

Gustave Dowd

Florida Gulf Coast University

gjdowd5096@eagle.fgcu.edu

April 2026

Abstract

Unaccounted-for water loss remains a persistent operational and financial challenge for municipal water utilities, particularly in rapidly growing communities where seasonal demand and infrastructure expansion can complicate system accountability. This study evaluated unaccounted-for water loss in the potable water system of Cape Coral using monthly utility records from January 2021 through December 2025. Monthly distributed water, billed consumption, estimated authorized unbilled use, customer connection counts, and precipitation data from the National Oceanic and Atmospheric Administration station at Southwest Florida International Airport were analyzed to quantify temporal variability and assess potential drivers of water loss. Unaccounted-for water loss was calculated as the difference between distributed water and the sum of billed water plus authorized unbilled water, expressed as both monthly volume and percentage of distributed water.

Annual unaccounted-for water loss ranged from 435 million gallons (MG) in 2022 to 705 MG in 2025, while monthly loss percentages ranged from approximately 0% to 24%. Seasonal variability was evident, with lower loss percentages generally occurring during late spring to early summer and higher recurring values during autumn months, particularly October and November. Simple linear regression identified a statistically significant inverse relationship between rainfall and monthly water loss percentage ($R^2 = 0.1636$, $p = 0.0013$) and a statistically significant positive relationship between distributed water volume and water loss percentage ($R^2 = 0.1032$, $p = 0.0123$). No statistically significant relationship was detected between active customer connections and monthly loss percentage ($R^2 = 0.0007$, $p = 0.8759$).

The results indicate that unaccounted-for water loss in Cape Coral was dynamic rather than static and was associated more strongly with seasonal demand conditions than with customer growth alone. These findings demonstrate the value of monthly data-driven water accountability programs and provide a practical framework for utilities seeking to improve efficiency, prioritize corrective actions, and strengthen long-term water resource management.

Introduction

Reliable potable water service depends not only on adequate source supply and treatment capacity, but also on the efficient delivery of finished water through distribution systems. A persistent challenge for utilities worldwide is the difference between water entering the system and water that is ultimately measured, billed, or otherwise accounted for through legitimate authorized uses. This difference is commonly described through terms such as non-revenue water, real losses, apparent losses, or unaccounted-for water depending on the accounting framework applied. Regardless of terminology, excessive losses reduce revenue recovery, increase treatment and pumping costs, accelerate infrastructure wear, and diminish long-term system resilience, particularly in communities experiencing rapid population growth and increasing service demands.

Water loss has been recognized as a major operational issue across both developed and developing water systems. The International Water Association (IWA) water balance framework separates losses into real losses, such as leakage from mains, storage tanks, and service connections, and apparent losses, such as meter inaccuracy, data handling errors, or unauthorized consumption (Lambert, 2002; Farley and Trow, 2003). The American Water Works Association (AWWA) has similarly promoted structured water auditing practices to improve utility accountability, benchmarking, and corrective action planning (AWWA, 2016). These frameworks emphasize that water loss is rarely attributable to a single cause; rather, it typically reflects the combined influence of infrastructure condition, metering performance, billing systems, operational practices, and customer demand behavior.

In many municipalities, water loss is reported as an annual summary percentage or annual audit value. While annual reporting is useful for regulatory compliance and long-term benchmarking, aggregated annual metrics may obscure shorter-term patterns that are operationally meaningful. Monthly or seasonal variation can reveal relationships with irrigation demand, tourism cycles, weather conditions, pressure changes, maintenance schedules, or billing system timing effects that are not evident in annual totals. Several studies have shown that higher temporal resolution data can improve leak detection, demand forecasting, and operational decision-making by identifying when system

inefficiencies emerge rather than simply quantifying yearly averages (Puust et al., 2010; Farley and Trow, 2003).

Climate and weather conditions are frequently linked to changing water demand, especially in warm regions where outdoor irrigation is common. In subtropical and arid environments, periods of lower rainfall often coincide with higher discretionary water use for landscaping and property maintenance. Increased demand during dry periods can elevate system pressures or prolong pumping durations, potentially increasing background leakage volumes and exposing metering discrepancies. Conversely, wetter months may reduce irrigation demand and lower total system throughput. Previous research has shown that weather-sensitive demand patterns can substantially affect utility production volumes and water balance outcomes (House-Peters and Chang, 2011; Balling and Gober, 2007). These dynamics are particularly relevant in Florida communities where seasonal precipitation contrasts and outdoor irrigation demand are pronounced.

Rapidly growing municipalities face additional complexity because system expansion can mask or compound water accountability challenges. New service connections, construction activity, meter installations, hydraulic changes, and expanding pipe networks can all influence loss metrics. However, growth does not necessarily translate directly into higher proportional losses. Utilities with strong asset management, accurate metering, and effective leak response may add customers while maintaining or improving accountability performance. As a result, customer growth should be evaluated empirically rather than assumed to be a primary driver of water loss (Kingdom et al., 2006).

Cape Coral, Florida provides an informative case study for examining these issues. Cape Coral is one of the fastest-growing cities in Florida and operates a large municipal potable water system supported by reverse osmosis treatment facilities and an extensive distribution network. Continued residential development, seasonal irrigation demand, and infrastructure expansion create operating conditions where understanding water accountability is increasingly important. Like many utilities, Cape Coral routinely tracks production and billing data; however, converting those records into actionable management insight requires structured analysis of temporal patterns and potential explanatory variables.

Despite widespread attention to water loss, comparatively fewer published case studies evaluate monthly operational trends using real utility records from rapidly growing coastal municipalities in the southeastern United States. Much of the existing literature focuses on annual audits, leakage technologies, or international systems with different infrastructure and climatic conditions. Accordingly, there remains practical value in applying established water balance methods to local utility datasets to determine whether recurring seasonal patterns or demand-related signals are present in a real municipal setting.

The purpose of this study was to evaluate unaccounted-for water loss in Cape Coral using monthly utility records from January 2021 through December 2025. Specifically, the study sought to (1) quantify annual and monthly variability in unaccounted-for water loss, (2) determine whether recurring seasonal patterns were evident, and (3) assess whether rainfall, distributed water demand, and customer connections were statistically associated with monthly loss percentage. Rather than proposing a novel theoretical model, this research applies established analytical methods in a high-growth municipal context where practical decision-making value is substantial.

The central hypothesis was that unaccounted-for water loss would vary systematically with seasonal demand conditions rather than being explained primarily by customer growth alone. If supported, such findings would suggest that utilities may benefit from shifting from static annual reporting toward monthly performance analytics that better identify when losses rise and when corrective actions may be most effective. As water utilities face increasing financial, environmental, and infrastructure pressures, improving accountability through data-driven management remains an essential component of sustainable urban water service.

Methods

This study evaluated unaccounted-for water loss in the municipal potable water system of Cape Coral using monthly operational records spanning January 2021 through December 2025 (N = 60 months). The methodological framework was developed from the combined perspectives of senior municipal utility practitioners, water loss specialists, environmental engineers, and quantitative analysts with extensive experience in applied water system performance assessment. The objective of the methods was to characterize

temporal variability in unaccounted-for water loss and examine whether selected operational and climatic variables were associated with observed changes in monthly loss percentages.

Study area and utility context

Cape Coral is a rapidly growing municipality in southwest Florida with an extensive potable water distribution network and a large customer base served by reverse osmosis water treatment facilities. Continued population growth, seasonal irrigation demand, infrastructure expansion, and weather variability make the system an appropriate case study for examining unaccounted-for water loss in a growing coastal utility. The study focused on system-level monthly records rather than individual pressure zones, meter districts, or neighborhood-scale assets.

Data sources

Monthly utility data were compiled from internal operational and billing records maintained by the City for the period January 2021 through December 2025. The dataset included monthly distributed water volume (million gallons, MG), billed water volume (MG), authorized unbilled water estimates (MG), and the number of active customer connections. Monthly rainfall totals were obtained from the National Oceanic and Atmospheric Administration (NOAA) station associated with Southwest Florida International Airport (RSW) in Fort Myers, Florida, which provides the nearest long-term publicly available climate record suitable for regional monthly precipitation analysis. Distributed water volume represented the monthly quantity of potable water introduced into the municipal distribution system. Billed water represented metered customer consumption recorded through the utility billing system. Monthly connection counts represented system snapshots of active customer accounts.

Authorized unbilled water estimation

Authorized unbilled water consisted of municipal water uses not directly billed to customers but considered legitimate system demand. These monthly estimates were developed from utility operational records and engineering judgment using available information on water main breaks, hydrant flushing, firefighting activity, system

maintenance operations, meter testing, operational drawdown events, construction-related water use, and other known municipal purposes. Because some categories were not continuously metered during the study period, estimated values were used where direct measurement was unavailable. These estimates were incorporated to reduce overstatement of true system loss.

Calculation of unaccounted-for water loss

Monthly unaccounted-for water loss was calculated as the difference between distributed water volume and the sum of billed water plus authorized unbilled water. Loss was expressed both as monthly volume (MG) and as a percentage of distributed water using Equation (1):

$$\text{Unaccounted-for Water Loss (\%)} = [(\text{Distributed} - (\text{Billed} + \text{Authorized Unbilled})) / \text{Distributed}] \times 100$$

This formulation was selected to better distinguish potential physical losses, accounting discrepancies, metering uncertainty, and other unexplained differences from known authorized municipal uses.

Data quality control and preprocessing

Records were reviewed for completeness, internal consistency, and abnormal values prior to analysis. No months were missing from the study period, and no interpolation was required. A known billing software transition affected January through March 2022. During that period, February 2022 was temporarily overstated relative to January and March because of a software-related billing adjustment. To minimize distortion in comparative analyses, the three monthly billed values for January, February, and March 2022 were replaced with their three-month average for this study. This smoothing approach preserved total quarterly magnitude while reducing a known administrative anomaly unrelated to actual water demand. All other monthly values were retained as originally recorded.

Statistical analysis

All analyses were conducted in Python using JupyterLab. Standard scientific computing libraries were used for data management, descriptive statistics, visualization, and regression analysis. Time-series plots were developed to evaluate long-term and month-to-month variation in distributed water, accounted-for water, and unaccounted-for water loss. Annual totals were aggregated to compare year-to-year differences. Monthly boxplots and a month-by-year heat map were developed to assess seasonal recurrence and intra-annual variability.

Bivariate relationships between monthly unaccounted-for water loss percentage and selected explanatory variables were evaluated using simple linear regression. Predictor variables included monthly rainfall, distributed water volume, and active customer connections. Separate univariate models were used to isolate the directional relationship of each predictor with loss percentage. For each regression, slope coefficients, coefficient of determination (R^2), and p-values were reported. An alpha significance threshold of 0.05 was adopted. Results with p-values less than 0.05 were considered statistically significant.

Variable selection and omitted factors

Air temperature was initially considered during project planning; however, complete temperature outputs were not incorporated into the final analytical dataset used for regression and therefore were excluded from inferential analyses. Additional potentially relevant drivers of water loss, such as system pressure fluctuations, leak repair response time, meter age, unauthorized consumption, and sub-annual tourism patterns, were not available in monthly standardized form and were outside the scope of the present dataset.

Reproducibility and limitations of method design

The analytical workflow was designed to be reproducible using standard monthly utility records and publicly available rainfall data. Another utility with comparable operational datasets could replicate the framework using the same equations and statistical procedures. Because the study relied on monthly aggregated data, the methods were intended to identify broad operational patterns rather than pinpoint specific leak locations or exact causes of loss events at daily or neighborhood scales.

Results

Monthly utility records from January 2021 through December 2025 (N = 60) were analyzed to evaluate unaccounted-for water loss in the City of Cape Coral. Unaccounted-for water loss was defined as the difference between distributed water and the sum of billed water plus estimated authorized unbilled water, expressed both as monthly volume (million gallons, MG) and as a percentage of distributed water. The analyses below address the study objectives of characterizing annual and monthly variability in unaccounted-for water loss and quantifying its bivariate relationships with rainfall, distributed demand, and number of customer connections.

Descriptive statistics and overall temporal patterns

Across the study period, monthly distributed water volumes generally ranged from approximately 355 to 500 MG, whereas billed plus authorized unbilled water ranged from approximately 285 to 463 MG (Figure 1). Distributed production exceeded accounted-for consumption throughout the study period, resulting in measurable unaccounted-for water loss in nearly all months.

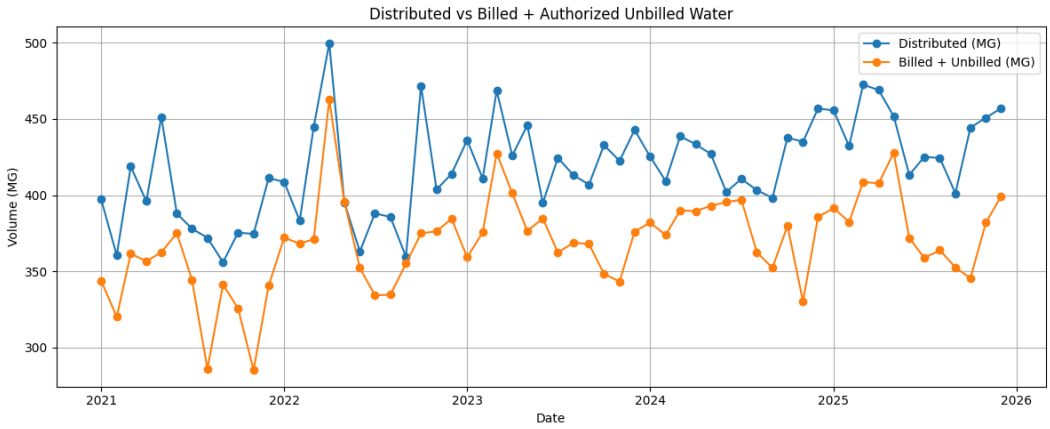

Figure 1. Distributed versus billed plus authorized unbilled water volumes over time (MG).

Annual unaccounted-for water loss volume varied substantially across the five-year record (Figure 2). Total annual unaccounted-for water loss was 635 MG in 2021,

decreased to 435 MG in 2022, increased to 633 MG in 2023, declined to 546 MG in 2024, and increased to 705 MG in 2025. Thus, the lowest annual loss occurred in 2022, whereas the highest occurred in 2025. The difference between these two years was 270 MG.

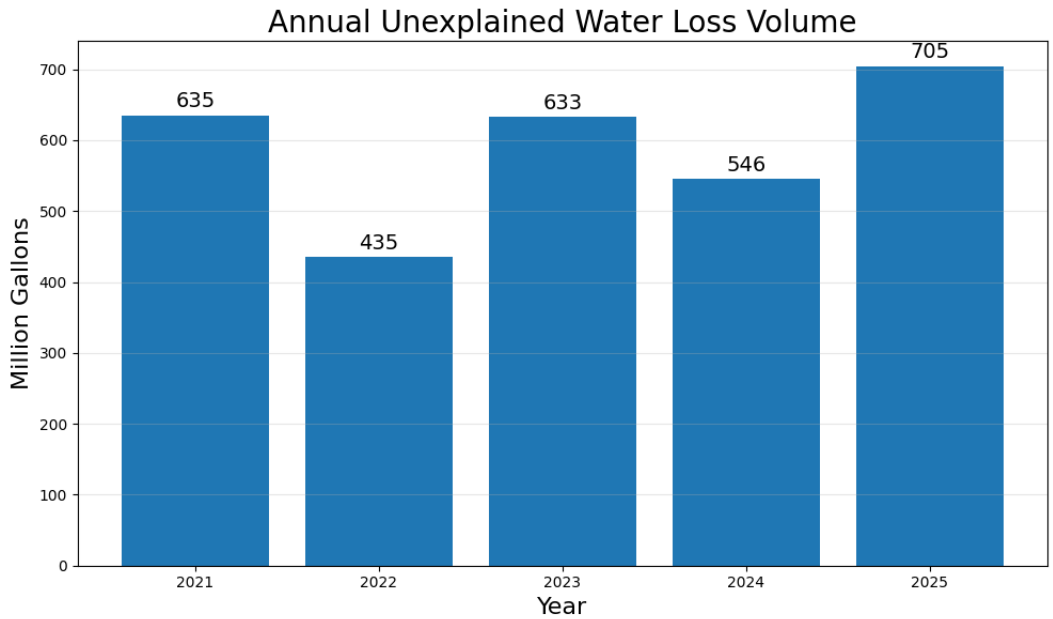

Figure 2. Annual unaccounted-for water loss volume (MG) by year.

Monthly unaccounted-for water loss volume also fluctuated markedly through time (Figure 3), ranging from near zero in isolated months to more than 100 MG during peak-loss months. Most monthly observations fell between approximately 25 and 75 MG. The corresponding percentage series showed similar variability (Figure 4). Monthly unaccounted-for water loss percentages ranged from approximately 0% to 24%, with most observations clustered between about 8% and 16%.

Taken together, these time-series results show that monthly water production remained within a comparatively narrow operating band relative to the larger variability observed in monthly unaccounted-for water loss percentage. This pattern was evident in both the volumetric and percentage expressions of loss.

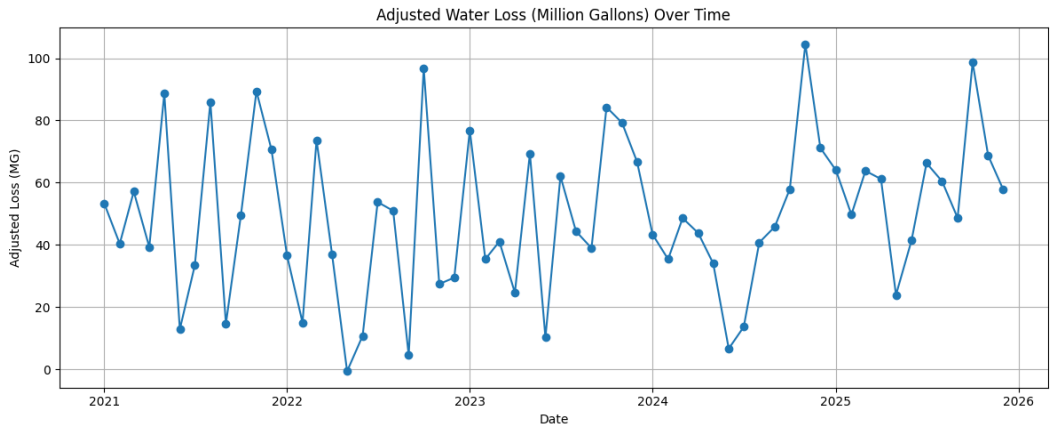

Figure 3. Monthly unaccounted-for water loss volume (MG) over time.

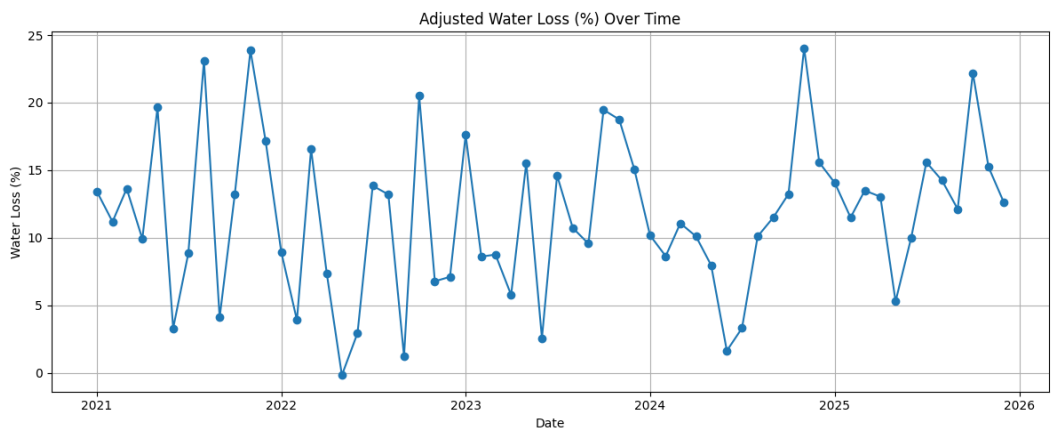

Figure 4. Monthly unaccounted-for water loss percentage over time.

Monthly and seasonal variation

Seasonal variation was evident in the distribution of monthly unaccounted-for water loss percentages (Figure 5). Median water loss percentages were lowest during early summer, particularly in June, where the median was approximately 3%. July values remained comparatively lower than many other months, although the spread was wider. By contrast, higher median losses were observed during October and November, with both months approaching or exceeding approximately 18%.

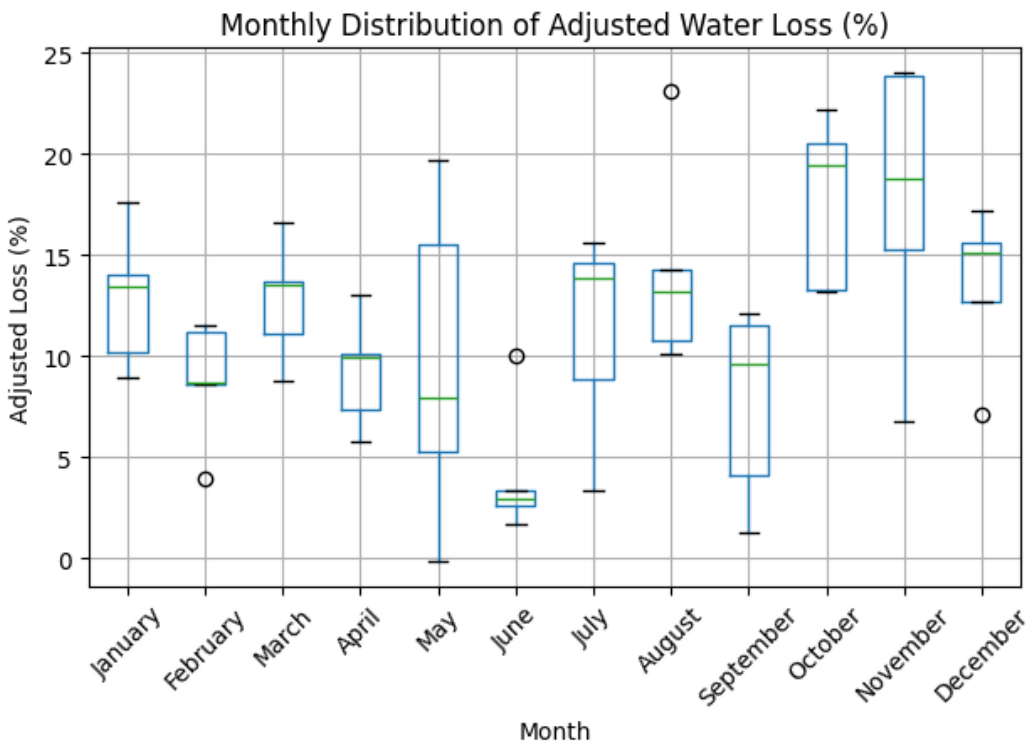

Figure 5. Monthly distribution of unaccounted-for water loss percentage showing seasonal variation.

The month-by-year heat map showed that this seasonal pattern recurred across multiple years rather than reflecting a single-year anomaly (Figure 6). Elevated unaccounted-for water loss percentages were repeatedly observed during autumn months, especially October and November. November exceeded 23% in both 2021 and 2024. In contrast, lower values were commonly observed during June and, in several years, May. Viewed across the full 2021–2025 record, fourth-quarter months (October–December) generally exhibited higher unaccounted-for water loss percentages than second-quarter months (April–June). This seasonal contrast was visible in both the boxplot distributions and the heat map and indicates recurring intra-annual variability in loss.

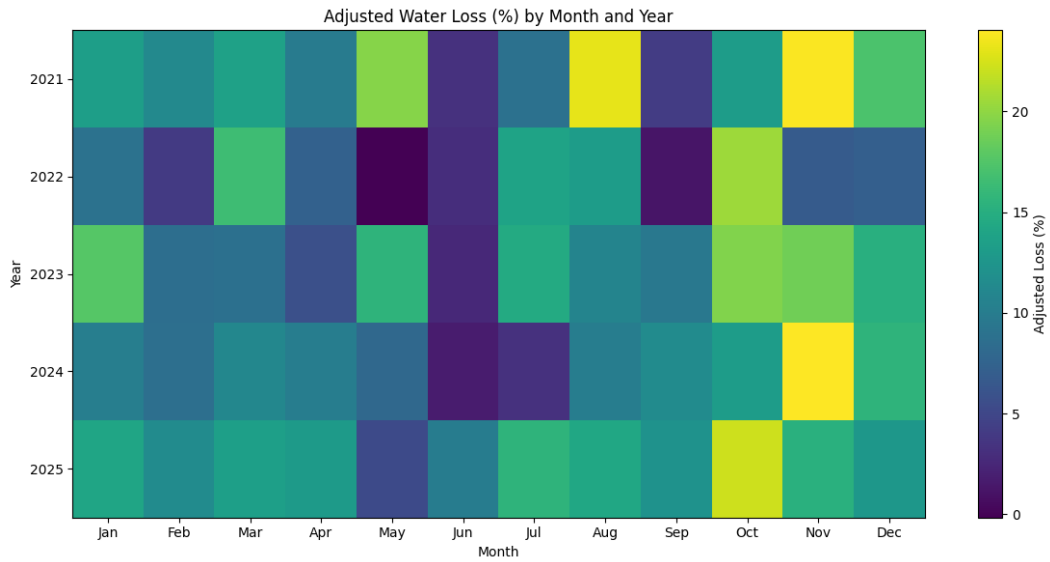

Figure 6. Heat map of monthly unaccounted-for water loss percentage by year and month.

Rainfall and unaccounted-for water loss

A statistically significant inverse relationship was observed between monthly rainfall and unaccounted-for water loss percentage (Figure 7). Simple linear regression produced a slope of -0.0585 percentage points of loss per inch of rainfall, with $R^2 = 0.1636$ and $p = 0.0013$. Under this fitted relationship, wetter months tended to correspond to lower unaccounted-for water loss percentages.

Among the univariate predictors evaluated in the present results set, rainfall had the highest explanatory power, accounting for 16.36% of the variance in monthly unaccounted-for water loss percentage. The scatter also showed substantial dispersion around the fitted line, indicating that rainfall alone did not account for most of the observed month-to-month variability.

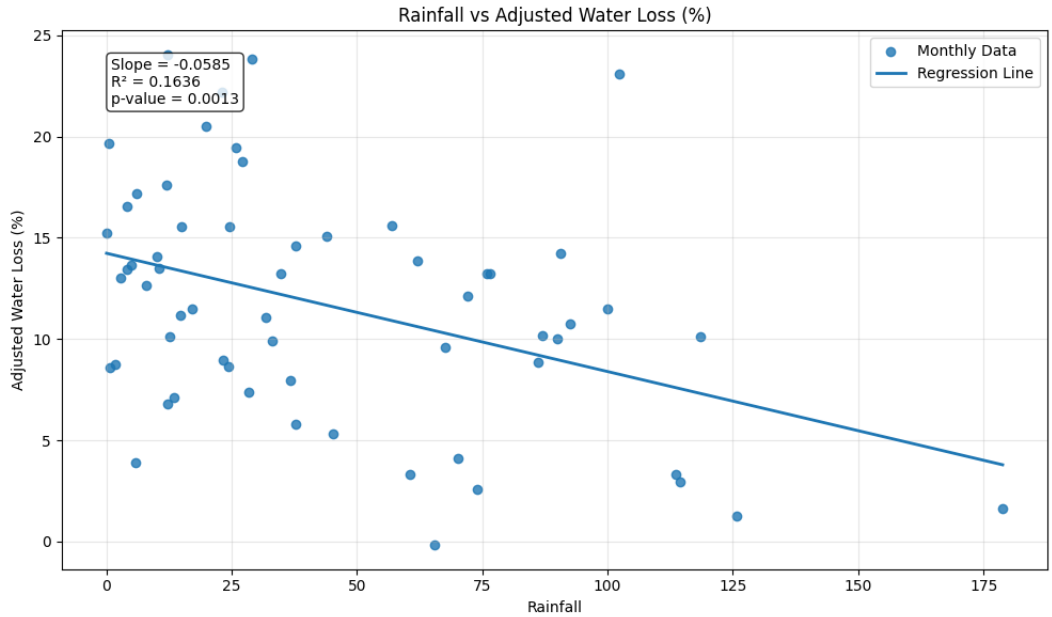

Figure 7. Relationship between monthly rainfall and unaccounted-for water loss percentage with regression line.

Distributed demand and unaccounted-for water loss

Monthly distributed water volume was positively associated with unaccounted-for water loss percentage (Figure 8). The fitted regression slope was 0.0586 percentage points of loss per MG distributed, with $R^2 = 0.1032$ and $p = 0.0123$. Months with higher distributed volumes therefore tended to exhibit higher proportional loss.

Although the relationship was statistically significant, the model explained 10.32% of the variance in monthly unaccounted-for water loss percentage, indicating modest explanatory strength in the univariate setting. The fitted trend was positive, but the data remained widely scattered around the regression line.

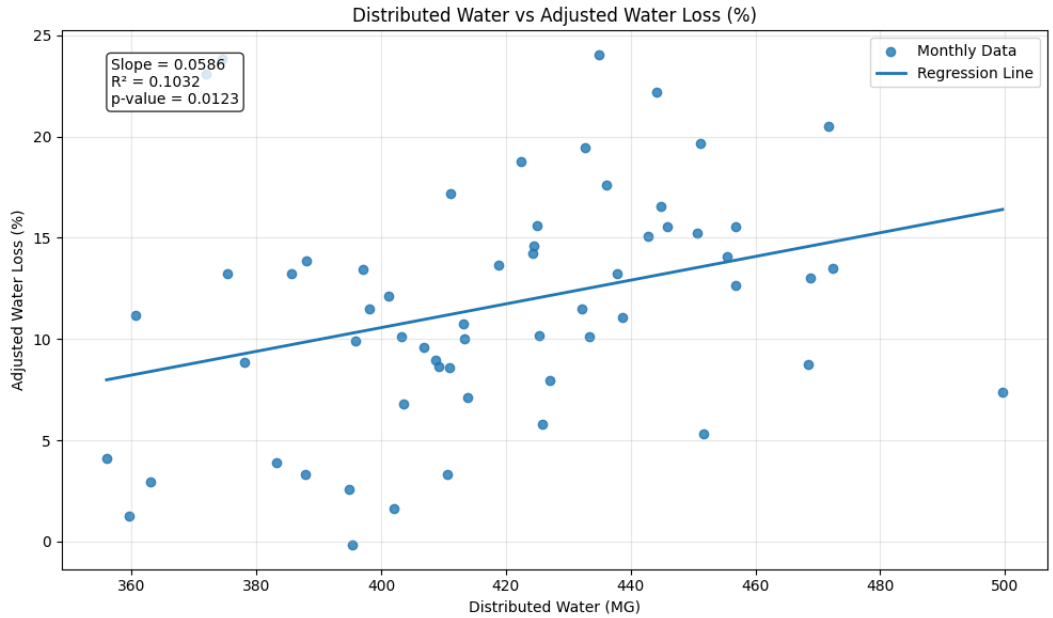

Figure 8. Relationship between distributed water volume and unaccounted-for water loss percentage with regression line.

Customer connections and unaccounted-for water loss

No statistically significant relationship was identified between the number of customer connections and unaccounted-for water loss percentage (Figure 9). The fitted slope was -0.000092, with $R^2 = 0.0007$ and $p = 0.8759$. The regression line was effectively flat, and the explanatory power was negligible.

The number of active connections increased through the study period, but that increase was not associated with detectable month-to-month changes in unaccounted-for water loss percentage in this bivariate analysis.

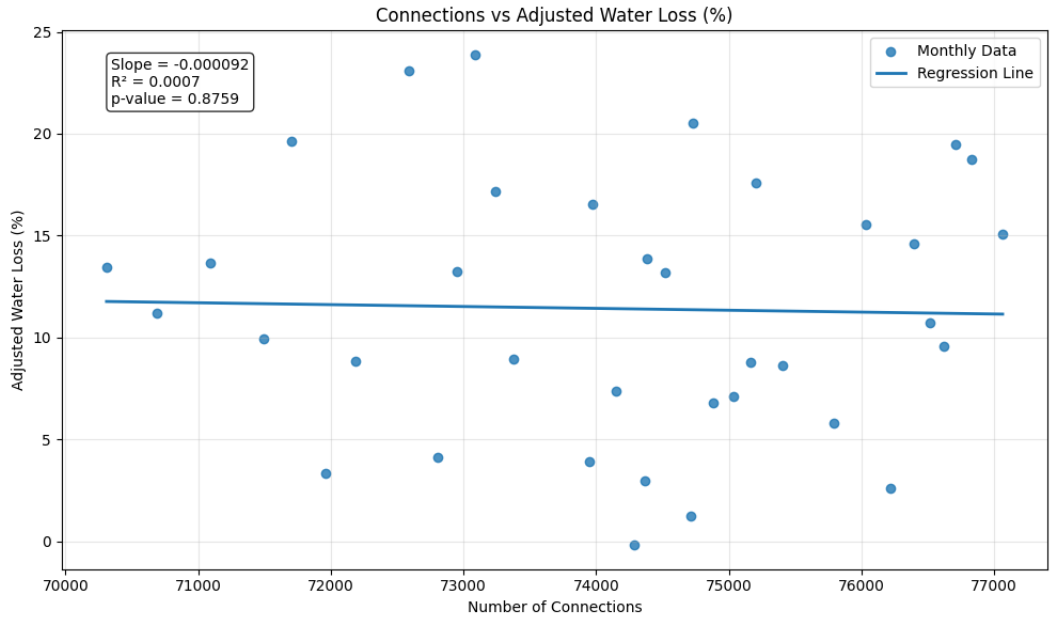

Figure 9. Relationship between number of customer connections and unaccounted-for water loss percentage with regression line.

Anomalies and outlying observations

Several monthly observations departed notably from the central range of the dataset. The highest monthly unaccounted-for water loss percentages approached 24% and occurred in late 2021 and late 2024 (Figures 4 and 6). Conversely, one month in 2022 produced a near-zero or slightly negative unaccounted-for water loss value, indicating that billed plus authorized unbilled water approximately equaled, or marginally exceeded, distributed water in that month.

The volumetric series similarly showed several high-loss months above 60 MG, particularly during 2025 (Figure 3). These elevated monthly observations contributed to the highest annual unaccounted-for water loss total in the record, which occurred in 2025 (Figure 2). In contrast, 2022 contained both the lowest annual total and several of the lowest monthly percentage values.

Summary of quantitative findings

Across the 2021–2025 study period, Cape Coral exhibited persistent but variable unaccounted-for water loss at both monthly and annual scales. Annual unaccounted-for water loss ranged from 435 to 705 MG, and monthly unaccounted-for water loss percentage ranged from approximately 0% to 24%. Seasonal variation was evident, with lower loss percentages generally occurring in late spring to early summer and higher percentages recurring during autumn, particularly October and November.

In the bivariate regression analyses, rainfall was negatively associated with unaccounted-for water loss percentage and distributed water volume was positively associated with unaccounted-for water loss percentage. Number of customer connections showed no detectable linear relationship with unaccounted-for water loss percentage in the present dataset. These results provide a quantitative description of the magnitude, timing, and basic correlations of unaccounted-for water loss over the five-year record.

Discussion

The present study evaluated unaccounted-for water loss in the potable water system of Cape Coral, Florida, using five years of monthly operational and billing records. The results demonstrate that unaccounted-for water loss was persistent throughout the study period but varied substantially across months and years rather than remaining constant. This finding is important because water loss is often discussed as a single annual percentage, whereas the present dataset shows that meaningful operational patterns may be masked when only annual summaries are considered. By examining monthly records, the analysis identified seasonal recurrence, statistically significant relationships with rainfall and distributed demand, and substantial interannual variability. Collectively, these findings indicate that unaccounted-for water loss in growing municipal systems is dynamic and influenced by both environmental and operational conditions.

One of the strongest quantitative findings was the inverse relationship between rainfall and unaccounted-for water loss percentage. Wetter months were generally associated with lower loss percentages, while drier months tended to correspond with higher losses. This pattern is consistent with operational expectations in subtropical municipalities where discretionary outdoor demand rises during dry periods. Landscape irrigation,

pressure washing, construction water use, and other seasonal uses often increase when rainfall is limited. Higher discretionary demand can increase total system throughput, elevate leakage volumes from existing defects, and magnify differences between produced and fully accounted-for water. In addition, some outdoor uses may be subject to metering inefficiencies or timing differences in billing cycles. Therefore, the rainfall relationship likely reflects both hydrologic seasonality and associated customer demand behavior rather than rainfall itself acting as a direct causal mechanism.

The positive association between distributed water volume and unaccounted-for water loss percentage further supports the importance of demand conditions. Months with higher distributed volumes tended to exhibit higher proportional losses. This relationship is consistent with principles of distribution system operation, where larger system throughput can increase leakage opportunity through joints, service lines, valves, and background seepage pathways. Periods of elevated production may also coincide with pump sequencing changes, longer run times, altered pressure zones, flushing events, or other operational responses that can influence water balance performance. Although the explanatory power of the simple regression model was modest, the consistency of the positive trend suggests that demand intensity is an important contributor to monthly loss variability.

The combined rainfall and distributed-volume findings are particularly informative because they point toward a recurring dry-season demand signal. Lower rainfall months commonly align with higher irrigation demand and higher total water production. Taken together, these two statistically significant relationships suggest that seasonal customer demand patterns may play a larger role in monthly unaccounted-for water loss than static infrastructure growth alone. For utility management, this means that loss reduction efforts may yield greater benefit when timed to seasonal demand peaks rather than applied uniformly throughout the year.

In contrast to the climatic and demand variables, the number of active customer connections showed no statistically significant relationship with monthly unaccounted-for water loss percentage. This result indicates that customer growth, by itself, did not explain short-term month-to-month changes in water loss over the study period. This finding is notable in the context of Cape Coral's continued development and system

expansion. Growth is often assumed to directly increase water loss because more customers require more infrastructure; however, the present results suggest that operational management, asset condition, metering performance, and seasonal demand may be more influential drivers of short-term variability than the simple count of connected accounts. Over longer time horizons, customer growth may still affect capital renewal needs and maintenance burden, but those effects were not detectable in the monthly bivariate analysis conducted here.

A prominent feature of the dataset was the recurring concentration of higher unaccounted-for water loss percentages during autumn months, particularly October and November. Because this pattern appeared across multiple years, it is unlikely to be explained solely by isolated anomalies. Several plausible mechanisms may contribute. First, residual outdoor demand may remain elevated after summer rainfall patterns weaken. Second, cumulative infrastructure stress from extended high-demand seasons may become more evident later in the year. Third, billing reconciliations, meter reading timing differences, or year-end administrative adjustments may cluster seasonally. Fourth, in southwest Florida, autumn overlaps with the latter portion of hurricane season, during which storm preparation, emergency operations, flushing, and repairs may influence water accounting. While the monthly dataset cannot isolate these mechanisms directly, the recurrence of autumn peaks strongly suggests that fourth-quarter performance review should be prioritized.

The observed interannual variability also has important management implications. Annual unaccounted-for water loss ranged from 435 MG to 705 MG during the five-year study period, representing a substantial difference in yearly performance. Such variability suggests that annual outcomes can shift meaningfully within relatively short operational timeframes. These differences may reflect changing climate conditions, evolving infrastructure condition, meter replacement cycles, operational practices, billing system refinements, or combinations of these factors. The lowest annual total occurred in 2022, while the highest occurred in 2025, indicating that performance improvements are possible but may not be sustained without continuous monitoring and responsive management.

From a utility operations perspective, the findings support a transition from static annual water loss reporting toward continuous monthly accountability management. Many utilities rely heavily on annual water audits to satisfy reporting requirements; however, annual metrics may not identify when losses occur or what conditions are associated with them. Monthly dashboards that track production, billed consumption, rainfall, seasonal trends, and unexplained deviations can provide earlier warning of emerging inefficiencies. Such systems can be paired with targeted leak detection, pressure management, meter testing, district metered area evaluation, and focused field investigations during historically high-loss months. In rapidly growing municipalities, this approach may provide greater operational value than annual retrospective summaries alone.

The study also has broader relevance beyond Cape Coral. Many Sun Belt and coastal communities experience similar combinations of rapid growth, strong seasonal irrigation demand, weather-sensitive consumption patterns, and expanding infrastructure networks. Utilities in these settings often face the same challenge of distinguishing true physical losses from metering uncertainty, authorized unbilled uses, and billing timing effects. The analytical framework used here—monthly water balance review combined with seasonal and regression analysis—offers a practical and transferable approach for other municipalities seeking to better understand unaccounted-for water loss using data already available in most utility information systems.

Study Limitations and Sources of Uncertainty

Several limitations should be considered when interpreting the results. First, the analysis relied on monthly aggregated data, which cannot identify the precise timing or geographic location of specific loss events. Short-duration leaks, pressure transients, unauthorized consumption incidents, or meter failures may be obscured when data are summarized monthly. Second, authorized unbilled water values included engineering estimates for some categories where direct metering was unavailable, introducing uncertainty into calculated unaccounted-for water loss values. Third, simple linear regression was intentionally used for interpretability, but univariate models do not capture interacting effects among rainfall, seasonality, demand, and operational changes. More advanced multivariate or time-series models may explain a larger share of

variability.

Fourth, rainfall data were obtained from the NOAA station at RSW Airport rather than multiple gauges distributed throughout Cape Coral. While regionally representative, localized storm variability may not be fully captured by a single station. Fifth, January through March 2022 required smoothing because of a documented billing software transition. This adjustment was appropriate to reduce administrative distortion, but any data modification introduces some analytical uncertainty. Finally, potentially relevant variables such as system pressure, infrastructure age, break frequency, meter age, tourism population shifts, and repair response times were not available in standardized monthly form.

Despite these limitations, the present study provides a robust five-year operational assessment based on real municipal utility records. The results demonstrate that unaccounted-for water loss was dynamic rather than static, showed recurring seasonal behavior, and was measurably associated with rainfall and system demand. These findings indicate that routine annual summaries alone may overlook actionable monthly patterns. For utilities experiencing continued growth, seasonal demand fluctuations, and expanding infrastructure obligations, a monthly data-driven accountability framework may provide greater value for identifying emerging inefficiencies and prioritizing corrective action. Accordingly, the approach developed in this study may be applicable not only to Cape Coral, but also to other rapidly developing municipalities seeking practical methods to better understand and reduce unaccounted-for water loss.

Conclusions

This study evaluated unaccounted-for water loss in the potable water system of Cape Coral using monthly operational, billing, and climatic records spanning January 2021 through December 2025. By applying a monthly water-balance framework rather than relying solely on annual summaries, the analysis identified persistent but highly variable unaccounted-for water loss across the five-year study period. The results demonstrate that system loss was not a static condition, but instead reflected recurring seasonal behavior, measurable environmental relationships, and meaningful year-to-year variation.

Several principal findings emerged from the analysis. First, annual unaccounted-for water

loss ranged from 435 MG to 705 MG, indicating substantial differences in yearly performance. Second, monthly unaccounted-for water loss percentages varied from approximately 0% to 24%, confirming that operational conditions changed materially across the study period. Third, a statistically significant inverse relationship was observed between rainfall and monthly unaccounted-for water loss percentage, while a statistically significant positive relationship was observed between distributed water volume and monthly unaccounted-for water loss percentage. These relationships indicate that dry-weather demand conditions and higher production months were associated with elevated losses. Fourth, the number of customer connections did not show a statistically detectable relationship with monthly unaccounted-for water loss percentage, suggesting that customer growth alone did not explain short-term variability.

The study also identified recurring seasonal concentration of higher losses during autumn months, particularly October and November. This repeated pattern across multiple years indicates that periodic operational, climatic, or administrative influences likely affect system accountability during the latter portion of the calendar year. As a result, utilities may benefit from targeted fourth-quarter reviews of production records, billing reconciliation, meter performance, leak detection priorities, and authorized unbilled uses. Concentrating resources during historically elevated loss periods may provide greater operational return than applying identical effort in all months.

From a management perspective, the findings support a transition from static compliance-oriented water loss reporting toward continuous operational analytics. Annual water audits remain valuable benchmarking tools; however, annual values alone may conceal the timing and magnitude of emerging inefficiencies. Monthly performance dashboards that integrate production, billed consumption, rainfall, and historical seasonal patterns can improve situational awareness and support proactive decision-making. Utilities can use such frameworks to prioritize leak investigations, optimize meter replacement programs, assess irrigation-season impacts, and refine operational controls.

The broader significance of this research extends beyond Cape Coral. Many rapidly growing municipalities in warm-climate regions face similar conditions, including seasonal irrigation demand, infrastructure expansion, fluctuating production volumes, and increasing pressure to improve water accountability. The analytical framework used in

this study relies largely on data already available in most utility systems and therefore offers a practical and transferable approach for other communities seeking to better quantify and manage unaccounted-for water loss.

Overall, the present study demonstrates that meaningful insight can be gained from routinely collected monthly utility records when analyzed systematically. Unaccounted-for water loss should be viewed not only as an annual reporting metric, but also as an operational performance indicator that changes through time and can be actively managed through targeted analysis, monitoring, and intervention.

Future Perspectives

Future research should expand beyond univariate monthly analysis to incorporate multivariate and time-series modeling frameworks that evaluate the combined effects of rainfall, demand, seasonality, billing cycles, and infrastructure condition. Inclusion of system pressure data, main break frequency, meter age, customer class consumption, irrigation restrictions, and repair response time may substantially improve explanatory power. Geographic analyses using pressure zones, district metered areas, or neighborhood-scale balances could help localize persistent loss patterns and improve field prioritization.

Additional work should also examine the financial implications of unaccounted-for water loss, including treatment cost, pumping energy, chemical use, and deferred capital impacts associated with excess production. Quantifying economic losses alongside volumetric losses would strengthen the business case for investment in metering modernization, leak detection technologies, and smart utility analytics.

Finally, future studies should evaluate whether management interventions informed by monthly analytics produce measurable reductions in water loss over time. Demonstrating successful operational improvement would help bridge the gap between academic analysis and utility practice, providing valuable guidance for Cape Coral and other municipalities pursuing long-term water efficiency and infrastructure resilience.

Data Availability Statement

The monthly operational data used in this study were derived from internal utility records maintained by the City of Cape Coral, Florida, including monthly distributed water volumes, billed consumption totals, estimated authorized unbilled water use, and active customer connections for the period January 2021 through December 2025.

Monthly precipitation data used in the study were obtained from the National Oceanic and Atmospheric Administration (NOAA) climate records associated with Southwest Florida International Airport (RSW), Fort Myers, Florida. These climatic records are publicly accessible through NOAA data repositories and associated federal climate databases.

Python scripts used for data cleaning, visualization, and statistical analyses can be recreated from the procedures described in the Methods section. If desired, code templates may be provided by the author upon request.

Ethics Statement

Utility data used in the analysis consisted of monthly system-level summaries of distributed water, billed consumption, estimated authorized unbilled use, and active customer connections. No customer-specific billing records, names, addresses, account identifiers, or individual consumption histories were accessed or analyzed. Data were evaluated only in aggregated monthly form for the purpose of assessing system-wide operational trends.

The study was conducted with the intent of improving municipal water accountability, operational efficiency, and long-term resource management. Findings are reported in good faith based on the records available and the analytical methods described in this manuscript. Where municipal records were used, access and use were assumed to be consistent with the author's professional responsibilities, applicable organizational policies, and relevant public-sector data governance practices.

Acknowledgments

The author gratefully acknowledges the City of Cape Coral Utilities Department and Water Production Division for maintaining the operational records that made this study possible. Appreciation is extended to utility personnel responsible for water production, distribution, billing administration, customer account management, field maintenance, instrumentation, and data reporting whose routine work contributed to the accuracy and continuity of the monthly records analyzed in this research.

The author also recognizes municipal staff involved in treatment plant operations, distribution system maintenance, meter management, leak response, and system planning. Their day-to-day efforts to sustain reliable potable water service under conditions of continued community growth provided the practical context for this study and informed the operational relevance of the research questions examined.

Special appreciation is extended to academic faculty, instructors, and mentors who provided guidance in engineering analysis, research methods, statistical interpretation, and technical writing during the development of this manuscript. Their instruction helped shape the analytical framework and presentation of results.

The author further acknowledges the National Oceanic and Atmospheric Administration (NOAA) for providing publicly accessible meteorological data used in the rainfall component of this analysis.

Finally, appreciation is extended to colleagues and reviewers who provided constructive discussion regarding utility performance metrics, water accountability practices, and manuscript organization. Any remaining errors, interpretations, or omissions are solely the responsibility of the author.

AI Assistance Statement

Artificial intelligence (AI) tools were used in a limited and transparent support capacity during the preparation of this manuscript. Specifically, AI-assisted language tools were

used to help improve sentence clarity, organization, grammar, formatting consistency, and draft efficiency for selected sections of the paper.

All research questions, study design decisions, data selection, analytical direction, interpretation of results, engineering judgment, and final conclusions were determined by the author. All numerical values, figures, tables, statistical outputs, and substantive technical content were reviewed and validated by the author prior to inclusion in the manuscript.

AI tools were not used as a substitute for independent analysis, professional judgment, or source verification. The author assumes full responsibility for the accuracy, integrity, originality, and final content of this work.

Funding Statement

This research received no external funding. No grants, sponsorships, contracts, or other direct financial support were provided by governmental agencies, private organizations, academic institutions, or commercial entities for the completion of this study. The work was conducted independently using existing municipal operational records and publicly available climatic data sources.

References

American Water Works Association (AWWA), 2016, M36 Water Audits and Loss Control Programs, 4th ed., American Water Works Association, Denver, Colo.

Balling, R. C., Jr., and P. Gober, 2007, Climate variability and residential water use in the city of Phoenix, Arizona: Journal of Applied Meteorology and Climatology, 46(7), 1130–1137, https://doi.org/10.1175/JAM2518.1

Farley, M., and S. Trow, 2003, Losses in Water Distribution Networks: A Practitioner's Guide to Assessment, Monitoring and Control, IWA Publishing, London, U.K.

House-Peters, L., and H. Chang, 2011, Urban water demand modeling: Review of concepts, methods, and organizing principles: Water Resources Research, 47, W05401, https://doi.org/10.1029/2010WR009624

Kingdom, B., R. Liemberger, and P. Marin, 2006, The Challenge of Reducing Non-Revenue Water (NRW) in Developing Countries: How the Private Sector Can Help, Water Supply and Sanitation Sector Board Discussion Paper Series No. 8, World Bank, Washington, D.C.

Lambert, A., 2002, International report: Water losses management and techniques: Water Science and Technology: Water Supply, 2(4), 1–20.

National Oceanic and Atmospheric Administration (NOAA), 2026, Climate data online: Monthly precipitation records for Southwest Florida International Airport (RSW), Fort Myers, Florida, National Centers for Environmental Information, accessed April 2026, https://www.ncei.noaa.gov/

Puust, R., Z. Kapelan, D. Savic, and T. Koppel, 2010, A review of methods for leakage management in pipe networks: Urban Water Journal, 7(1), 25–45, https://doi.org/10.1080/15730620903447665

Thornton, J., R. Sturm, and G. Kunkel, 2008, Water Loss Control, 2nd ed., McGraw-Hill, New York.

United States Environmental Protection Agency (U.S. EPA), 2013, Water Audits and Water Loss Control for Public Water Systems, EPA 816-F-13-002, U.S. Environmental Protection Agency, Washington, D.C.

World Bank, 2018, The Role of Performance Data in Reducing Non-Revenue Water in Urban Utilities, World Bank Group, Washington, D.C.